

**Supplemental Material for “Two spectral-resource regimes for
autonomous temporal information”**

Anonymous

Anonymous

This Supplemental Material gives self-contained proof cores for the resource statements used in the main text. We work with finite-dimensional clock and signal Hilbert spaces, as appropriate to the finite-internal-system formulation studied here; infinite-dimensional extensions require separate domain and closed-range hypotheses and are not claimed. The order follows the physical hierarchy: finite-radius survival, rank-changing boundary synthesis, the arbitrary-support mixed bridge, the multi-gap shared-Hessian law, and finally the exact qutrit common-record accessibility benchmark. Standard ingredients—numerical-radius inequalities, PSD-cone second-order geometry, shorted operators, and multiparameter Fisher bounds—are used as mathematical infrastructure rather than novelty claims [1–6].

I. FINITE-RADIUS SURVIVAL

A. Finite-radius arbitrary-POVM survival laws

We use the convention

$$D_c = (A + A^\dagger)/2, \quad D_s = (A - A^\dagger)/(2i). \quad (1)$$

Let $P = \text{supp } \rho_0$ and suppose the affine disk

$$\rho_0 + xD_c + yD_s \succeq 0 \quad (x^2 + y^2 \leq R^2) \quad (2)$$

is physical for some $R > 0$. Then positivity in both signs implies $A = PAP$.

Define on P

$$B = \rho_0^{-1/2} A \rho_0^{-1/2}. \quad (3)$$

Writing $\beta = x - iy$, the affine family is congruent to

$$\mathbb{I} + \frac{1}{2}(\beta B + \beta^* B^\dagger). \quad (4)$$

For a unit vector $|\psi\rangle$, minimizing the expectation value over the phase of β gives

$$1 - |\beta| |\langle \psi | B | \psi \rangle|. \quad (5)$$

Therefore the exact affine radius is

$$R_{\text{lin}} = \frac{1}{w(B)}. \quad (6)$$

The standard numerical-radius inequality $\|B\| \leq 2w(B)$ gives

$$BB^\dagger \preceq \frac{4}{R_{\text{lin}}^2} \mathbb{I}_P. \quad (7)$$

By congruence,

$$A\rho_0^+A^\dagger = \rho_0^{1/2}BB^\dagger\rho_0^{1/2} \preceq \frac{4}{R_{\text{lin}}^2}\rho_0. \quad (8)$$

Similarly,

$$A^\dagger\rho_0^+A \preceq \frac{4}{R_{\text{lin}}^2}\rho_0. \quad (9)$$

For any POVM $\{M_y\}$ define

$$p_y = \text{Tr}(\rho_0 M_y), \quad z_y = \text{Tr}(A M_y). \quad (10)$$

Weighted Hilbert–Schmidt Cauchy–Schwarz gives

$$\frac{|z_y|^2}{p_y} \leq \text{Tr}(M_y A \rho_0^+ A^\dagger). \quad (11)$$

After summing the outcomes,

$$\text{Tr } F_1 \leq \text{Tr}(A \rho_0^+ A^\dagger). \quad (12)$$

Now suppose the range of A lies in a projector P_U : $P_U A = A$. The positive operator on the left of Eq. (8) is then supported in P_U , hence

$$\text{Tr } F_1 \leq \text{Tr}(A \rho_0^+ A^\dagger) \quad (13)$$

$$= \text{Tr}[P_U(A \rho_0^+ A^\dagger)] \quad (14)$$

$$\leq \frac{4}{R_{\text{lin}}^2} \text{Tr}(P_U \rho_0). \quad (15)$$

Thus

$$\frac{R_{\text{lin}}^2}{4} \text{Tr } F_1 \leq \text{Tr}(P_U \rho_0). \quad (16)$$

The same argument applied to $A^\dagger$ yields a domain/lower-endpoint bound.

a. Finite-copy extension. For N independently encoded copies,

$$A_N = \sum_{j=1}^N \rho_0^{\otimes(j-1)} \otimes A \otimes \rho_0^{\otimes(N-j)}. \quad (17)$$

Because $\text{Tr } A = 0$, all cross-copy terms in

$$\text{Tr}(A_N \rho_N^+ A_N^\dagger) \quad (18)$$

vanish, while every diagonal copy term equals the one-copy weighted norm. Therefore

$$\text{Tr}(A_N \rho_N^\dagger A_N^\dagger) = N \text{Tr}(A \rho_0^\dagger A^\dagger). \quad (19)$$

The weighted score inequality is valid for an arbitrary POVM on the full N -copy Hilbert space, so

$$\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N}{N} \leq \text{Tr}(P_U \rho_0). \quad (20)$$

No product-measurement assumption is used.

b. Exact Bohr mode. If

$$[H, A_\nu] = \hbar \nu A_\nu, \quad (21)$$

then the range and domain of A_ν lie in upper and lower endpoint sectors separated by $\hbar \nu$. For a stationary baseline, applying Eq. (20) to A_ν and $A_\nu^\dagger$ gives

$$\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq \min\{U_\nu, D_\nu\}. \quad (22)$$

For a semibounded Hamiltonian the upper endpoint lies inside the tail

$$T(\nu) = \Pr_{\rho_0}[H - E_* \geq \hbar \nu], \quad (23)$$

so

$$\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq T(\nu). \quad (24)$$

Markov's inequality then gives the mean-energy corollary.

B. Autonomous dual survival

For the exact clock-signal exchange

$$[H_S, A_\nu] = +\hbar \nu A_\nu, \quad [H_C, A_\nu] = -\hbar \nu A_\nu, \quad (25)$$

the signal viewpoint sees A_ν as a positive $+\nu$ tangent, whereas the clock viewpoint sees $A_\nu^\dagger$ as a positive $+\nu$ tangent. The two Hermitian tangent quadratures span the same physical plane under $A_\nu \leftrightarrow A_\nu^\dagger$; only the sine coordinate reverses. Hence the affine radius and Fisher trace are unchanged.

Applying Eq. (20) to the signal range projector and to the clock range projector of $A_\nu^\dagger$ yields

$$\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq T_S(\nu), \quad (26)$$

and

$$\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq T_C(\nu). \quad (27)$$

Therefore

$$\boxed{\frac{R_{\text{lin}}^2}{4} \frac{\text{Tr } F_N^{(\nu)}}{N} \leq \min\{T_C(\nu), T_S(\nu)\}.} \quad (28)$$

The proof requires only global physicality of the baseline and exact local gap structure of the tangent; separate local stationarity is unnecessary because Eq. (20) did not assume $[\rho_0, H_X] = 0$.

II. RANK-CHANGING BOUNDARY SYNTHESIS AND AUTONOMOUS ACTION

A. Rank-changing boundary curvature and autonomous action

Let $\rho(x, y)$ be a C^2 physical family with baseline ρ_0 , support projector $P = \text{supp } \rho_0$, and kernel projector $Q = \mathbb{I} - P$. Let

$$\partial_x \rho(0) = D_c = \frac{A + A^\dagger}{2}, \quad \partial_y \rho(0) = D_s = \frac{A - A^\dagger}{2i}. \quad (29)$$

1. One-sided boundary synthesis

Assume a baseline-empty endpoint projector P_U with

$$A = P_U A P, \quad P_U P = 0. \quad (30)$$

Define

$$J = \text{Tr}(A \rho_0^+ A^\dagger), \quad T_U(x, y) = \text{Tr}[P_U \rho(x, y)]. \quad (31)$$

For any POVM, the right-supported weighted score inequality gives

$$\text{Tr } F_1 \leq J. \quad (32)$$

The independent-copy identity gives

$$\frac{\text{Tr } F_N}{N} \leq J \quad (33)$$

for arbitrary collective measurements.

To obtain the curvature cost, apply the second-order PSD-cone condition separately to the x and y curves. Since

$$QD_cP = \frac{A}{2}, \quad QD_sP = \frac{A}{2i}, \quad (34)$$

one has

$$Q\partial_x^2\rho(0)Q \succeq 2\left(\frac{A}{2}\right)\rho_0^+\left(\frac{A}{2}\right)^\dagger = \frac{1}{2}A\rho_0^+A^\dagger, \quad (35)$$

$$Q\partial_y^2\rho(0)Q \succeq 2\left(\frac{A}{2i}\right)\rho_0^+\left(\frac{A}{2i}\right)^\dagger = \frac{1}{2}A\rho_0^+A^\dagger. \quad (36)$$

Adding gives

$$C_\Delta := Q(\partial_x^2 + \partial_y^2)\rho(0)Q \succeq A\rho_0^+A^\dagger. \quad (37)$$

Taking the trace after projection onto P_U yields

$$\Delta T_U(0) \geq J. \quad (38)$$

Thus

$$\boxed{\frac{\text{Tr } F_N}{N} \leq J \leq \Delta T_U(0).} \quad (39)$$

2. Bilateral boundary Minkowski law

Now assume a pure-boundary tangent

$$PAP = 0, \quad QAQ = 0. \quad (40)$$

Define

$$X = QAP, \quad Y = QA^\dagger P. \quad (41)$$

Then

$$A = X + Y^\dagger. \quad (42)$$

For a POVM define

$$u_y = \frac{\text{Tr}(XM_y)}{\sqrt{p_y}}, \quad v_y = \frac{\text{Tr}(YM_y)^*}{\sqrt{p_y}}. \quad (43)$$

The Fisher trace is the squared norm of the sum of score vectors,

$$\text{Tr } F_1 = \|u + v\|_2^2. \quad (44)$$

Weighted Cauchy–Schwarz gives

$$\|u\|_2^2 \leq J_+, \quad \|v\|_2^2 \leq J_-, \quad (45)$$

where

$$J_+ = \text{Tr}(X\rho_0^+X^\dagger), \quad J_- = \text{Tr}(Y\rho_0^+Y^\dagger). \quad (46)$$

Therefore Minkowski gives

$$\boxed{\sqrt{\text{Tr } F_1} \leq \sqrt{J_+} + \sqrt{J_-}.} \quad (47)$$

For N copies the two weighted norms scale as NJ_+ and NJ_- , hence

$$\boxed{\sqrt{\frac{\text{Tr } F_N}{N}} \leq \sqrt{J_+} + \sqrt{J_-}.} \quad (48)$$

For the second-order resource, the support-to-kernel blocks of the two real derivatives are

$$K_c = \frac{X+Y}{2}, \quad K_s = \frac{X-Y}{2i}. \quad (49)$$

The two PSD-cone inequalities give

$$Q\partial_x^2\rho Q \succeq 2K_c\rho_0^+K_c^\dagger, \quad (50)$$

$$Q\partial_y^2\rho Q \succeq 2K_s\rho_0^+K_s^\dagger. \quad (51)$$

Adding them cancels the X/Y cross terms exactly:

$$C_\Delta \succeq X\rho_0^+X^\dagger + Y\rho_0^+Y^\dagger = Z_+ + Z_-. \quad (52)$$

If X and Y enter orthogonal baseline-empty endpoint sectors P_+ and P_- , define

$$T_\pm(x, y) = \text{Tr}(P_\pm\rho(x, y)). \quad (53)$$

Projecting Eq. (52) onto the two sectors gives

$$J_+ \leq \Delta T_+(0), \quad J_- \leq \Delta T_-(0). \quad (54)$$

Combining with Eq. (48),

$$\frac{\text{Tr } F_N}{N} \leq \left[\sqrt{\Delta T_+(0)} + \sqrt{\Delta T_-(0)} \right]^2. \quad (55)$$

The cross term in the square is generally physical and cannot be replaced by an additive curvature bound.

3. Autonomous two-sided action

Assume now the exact exchange relations

$$[H_S, A_\nu] = +\hbar\nu A_\nu, \quad [H_C, A_\nu] = -\hbar\nu A_\nu. \quad (56)$$

In a clean bilateral boundary geometry, X creates a signal-upper and clock-lower endpoint, whereas Y creates a signal-lower and clock-upper endpoint. Let the corresponding local population Laplacians be

$$\Delta T_{S,+}, \quad \Delta T_{S,-}, \quad \Delta T_{C,+}, \quad \Delta T_{C,-}. \quad (57)$$

Define

$$\mathcal{A}_S^{(2)} = \frac{\hbar\nu}{4}(\Delta T_{S,+} + \Delta T_{S,-}), \quad (58)$$

$$\mathcal{A}_C^{(2)} = \frac{\hbar\nu}{4}(\Delta T_{C,+} + \Delta T_{C,-}). \quad (59)$$

From Eq. (55) and

$$(\sqrt{a} + \sqrt{b})^2 \leq 2(a + b), \quad (60)$$

each subsystem obeys

$$\mathcal{A}_X^{(2)} \geq \frac{\hbar\nu}{8} \frac{\text{Tr } F_N}{N}, \quad X \in \{C, S\}. \quad (61)$$

Therefore

$$\boxed{\mathcal{A}_C^{(2)} + \mathcal{A}_S^{(2)} \geq \frac{\hbar\nu}{4} \frac{\text{Tr } F_N}{N}.} \quad (62)$$

If only one orientation is present, Eq. (39) applies on each local side separately. Each subsystem then obeys

$$\mathcal{A}_X^{(2)} \geq \frac{\hbar\nu}{4} \frac{\text{Tr } F_N}{N}, \quad (63)$$

so

$$\boxed{\mathcal{A}_C^{(2)} + \mathcal{A}_S^{(2)} \geq \frac{\hbar\nu}{2} \frac{\text{Tr } F_N}{N}.} \quad (64)$$

4. Sharp fixed-shell constructions

For bilateral synthesis use the total-excitation-2 shell

$$|L\rangle = |2_C, 0_S\rangle, \quad |M\rangle = |1_C, 1_S\rangle, \quad |U\rangle = |0_C, 2_S\rangle, \quad (65)$$

with

$$A_\nu = c(|U\rangle\langle M| + |M\rangle\langle L|), \quad \rho_0 = |M\rangle\langle M|. \quad (66)$$

The exact family

$$|\psi(x, y)\rangle = \sqrt{1 - \frac{c^2}{2}(x^2 + y^2)}|M\rangle \quad (67)$$

$$+ \frac{c}{2}(x + iy)|L\rangle + \frac{c}{2}(x - iy)|U\rangle \quad (68)$$

is normalized on the open disk $x^2 + y^2 < 2/c^2$ and lies entirely inside one total-energy eigenspace. Every local endpoint Laplacian is c^2 , while the three-outcome Fourier measurement gives

$$\text{Tr } F_1 = 4c^2. \quad (69)$$

Thus Eq. (62) is saturated exactly.

For one-sided synthesis use

$$A_\nu = 2c|U\rangle\langle D|, \quad \rho_0 = |D\rangle\langle D|, \quad (70)$$

inside a fixed-total-excitation-1 shell and the convention-consistent family

$$|\psi(x, y)\rangle = \sqrt{1 - c^2(x^2 + y^2)}|D\rangle + c(x - iy)|U\rangle. \quad (71)$$

The local endpoint Laplacians are $4c^2$, a fixed equatorial POVM gives $\text{Tr } F_1 = 4c^2$, and Eq. (64) is saturated exactly.

III. ARBITRARY COHERENT SUPPORT

A. Arbitrary coherent-support mixed survival-synthesis bridge

All statements in this subsection are finite-dimensional. Let A_ν satisfy

$$[H_S, A_\nu] = +\hbar\nu A_\nu, \quad [H_C, A_\nu] = -\hbar\nu A_\nu, \quad (72)$$

and let $P = \text{supp } \rho_0$, $Q = \mathbb{I} - P$. Two-sided first-order physicality implies

$$QA_\nu Q = 0. \quad (73)$$

Decompose

$$B = PA_\nu P, \quad K_+ = QA_\nu P, \quad K_- = QA_\nu^\dagger P, \quad (74)$$

so $A_\nu = B + K_+ + K_-^\dagger$. Define

$$J_B^+ = \text{Tr}(B\rho_0^+B^\dagger), \quad J_B^- = \text{Tr}(B^\dagger\rho_0^+B), \quad (75)$$

$$J_+ = \text{Tr}(K_+\rho_0^+K_+^\dagger), \quad J_- = \text{Tr}(K_-\rho_0^+K_-^\dagger). \quad (76)$$

1. Measurement-side master inequality

Set $X = (B + K_+)P$ and $Y = K_-$. The outputs of B and K_+ lie in orthogonal P and Q sectors, hence

$$\text{Tr}(X\rho_0^+X^\dagger) = J_B^+ + J_+. \quad (77)$$

Weighted Hilbert–Schmidt Cauchy–Schwarz and score-space Minkowski give

$$\sqrt{\text{Tr } F_1} \leq \sqrt{J_B^+ + J_+} + \sqrt{J_-}. \quad (78)$$

Applying the same argument to $A_\nu^\dagger$ gives

$$\sqrt{\text{Tr } F_1} \leq \sqrt{J_B^- + J_-} + \sqrt{J_+}. \quad (79)$$

All weighted norms scale exactly by N for independent encoding, so for every finite N and arbitrary collective POVM,

$$\boxed{\sqrt{\frac{\text{Tr } F_N}{N}} \leq \min\left\{\sqrt{J_B^+ + J_+} + \sqrt{J_-}, \sqrt{J_B^- + J_-} + \sqrt{J_+}\right\}. \quad (80)}$$

2. Two-sided shorted survival ceilings

Because P need not commute with either local Hamiltonian, $B = PAP$ need not itself be an exact Bohr-gap operator. Let

$$Z_B^+ = B\rho_0^+B^\dagger, \quad Z_B^- = B^\dagger\rho_0^+B, \quad (81)$$

with support projectors $R_B^+ = \text{supp } Z_B^+$ and $R_B^- = \text{supp } Z_B^-$. Let $\Pi_{S,U}, \Pi_{S,D}, \Pi_{C,U}, \Pi_{C,D}$ be minimal participating endpoint projectors for the support-preserving exchange content and define

$$S_{S,U} = P\Pi_{S,U}P, \quad S_{S,D} = P\Pi_{S,D}P, \quad (82)$$

$$S_{C,U} = P\Pi_{C,U}P, \quad S_{C,D} = P\Pi_{C,D}P. \quad (83)$$

Define

$$\lambda_{S,U} = \sup\{\lambda \geq 0 : S_{S,U} \succeq \lambda R_B^+\}, \quad (84)$$

$$\lambda_{C,D} = \sup\{\lambda \geq 0 : S_{C,D} \succeq \lambda R_B^+\}, \quad (85)$$

$$\lambda_{S,D} = \sup\{\lambda \geq 0 : S_{S,D} \succeq \lambda R_B^-\}, \quad (86)$$

$$\lambda_{C,U} = \sup\{\lambda \geq 0 : S_{C,U} \succeq \lambda R_B^-\}. \quad (87)$$

Let R_B be the affine radius of the support-preserving sub-tangent and $T_{X,E} = \text{Tr}(\Pi_{X,E}\rho_0)$.

Whenever the relevant shorting constant is positive, the finite-radius majorization gives

$$J_B^+ \leq \frac{4T_{S,U}}{R_B^2 \lambda_{S,U}}, \quad J_B^+ \leq \frac{4T_{C,D}}{R_B^2 \lambda_{C,D}}, \quad (88)$$

$$J_B^- \leq \frac{4T_{S,D}}{R_B^2 \lambda_{S,D}}, \quad J_B^- \leq \frac{4T_{C,U}}{R_B^2 \lambda_{C,U}}. \quad (89)$$

A local-side bound with $\lambda = 0$ is simply unavailable; no extended-real value is inserted into later formulas. For $B \neq 0$, define a_+ and a_- as the minima over the available finite local-side ceilings for their respective orientations. If neither local side has a positive shorting constant for an orientation, declare that scalar internal-resource branch unavailable. If $B = 0$, define both branches and set

$$a_+ = a_- = 0. \quad (90)$$

Whenever a branch is available,

$$J_B^+ \leq a_+ \quad \text{or} \quad J_B^- \leq a_- \quad (91)$$

as appropriate.

3. Canonical endpoint-incidence action

Define

$$\Pi_{\text{out}} = \text{supp}(A_\nu A_\nu^\dagger), \quad \Pi_{\text{in}} = \text{supp}(A_\nu^\dagger A_\nu). \quad (92)$$

The exact exchange relations imply that both commute with H_C and H_S . Define

$$G_{\text{ex}} = 2\hbar\nu Q(\Pi_{\text{out}} + \Pi_{\text{in}})Q. \quad (93)$$

For the kernel Laplacian

$$C_\Delta = Q(\partial_x^2 + \partial_y^2)\rho(0)Q, \quad (94)$$

second-order positivity gives

$$C_\Delta \succeq Z_+ + Z_-, \quad Z_+ = K_+ \rho_0^+ K_+^\dagger, \quad Z_- = K_- \rho_0^+ K_-^\dagger. \quad (95)$$

Define

$$\mathcal{A}_{\text{ex}}^{(2)} = \frac{1}{4} \text{Tr}(G_{\text{ex}} C_\Delta), \quad e := 4\mathcal{A}_{\text{ex}}^{(2)}. \quad (96)$$

For each nonzero synthesized orientation define $R_\pm = \text{supp } Z_\pm$ and

$$g_\pm = \lambda_{\min}(R_\pm G_{\text{ex}} R_\pm |_{R_\pm}). \quad (97)$$

Whenever all nonzero synthesized ranges have positive prices,

$$g_+ J_+ + g_- J_- \leq e, \quad (98)$$

with absent orientations omitted. If a nonzero orientation has $g_\pm = 0$, no finite scalar action-only theorem can bound its $J_\pm$; one must retain a different cost operator or the operator-valued curvature itself.

4. Bilateral scalar envelope

For $a, e \geq 0$ and $p, q > 0$, define

$$\Psi_a(e; p, q) = \begin{cases} (\sqrt{a} + \sqrt{e/q})^2, & e \leq ap^2/q, \\ (e + pa)(p^{-1} + q^{-1}), & e \geq ap^2/q. \end{cases} \quad (99)$$

This is the exact maximum of

$$[\sqrt{a + j_+} + \sqrt{j_-}]^2 \quad (100)$$

subject to $pj_+ + qj_- \leq e$, $j_\pm \geq 0$.

If both synthesized orientations are nonzero, $g_+, g_- > 0$, and both internal branches are available, Eqs. (80)–(98) give

$$\boxed{\frac{\text{Tr } F_N}{N} \leq \min\{\Psi_{a_+}(e; g_+, g_-), \Psi_{a_-}(e; g_-, g_+)\}}. \quad (101)$$

If only one internal branch is available, retain only the corresponding Ψ term. If neither is available, the scalar mixed reduction is vacuous even though the operator/measurement inequalities remain valid.

5. Degenerate synthesis orientations

Absent synthesized ranges are handled directly from Eq. (80); no restricted eigenvalue is assigned to an absent range.

If $J_- = 0$, $J_+ > 0$, and $g_+ > 0$, the two possible finite branches are

$$\frac{\text{Tr } F_N}{N} \leq a_+ + \frac{e}{g_+} \quad (102)$$

when a_+ is available, and

$$\frac{\text{Tr } F_N}{N} \leq \left(\sqrt{a_-} + \sqrt{\frac{e}{g_+}} \right)^2 \quad (103)$$

when a_- is available. Take the minimum over the available branches. If neither internal branch is available, no finite scalar mixed bound follows.

If $J_+ = 0$, $J_- > 0$, and $g_- > 0$, the available branches are

$$\frac{\text{Tr } F_N}{N} \leq \left(\sqrt{a_+} + \sqrt{\frac{e}{g_-}} \right)^2 \quad (104)$$

when a_+ is available, and

$$\frac{\text{Tr } F_N}{N} \leq a_- + \frac{e}{g_-} \quad (105)$$

when a_- is available. Again take the minimum over available branches.

If $J_+ = J_- = 0$, each available internal branch independently gives

$$\frac{\text{Tr } F_N}{N} \leq a_+ \quad \text{or} \quad \frac{\text{Tr } F_N}{N} \leq a_-. \quad (106)$$

Take their minimum when both are available; if neither is available, the scalar survival reduction is vacuous.

In the clean pure-boundary plus-only or minus-only case, $a_+ = a_- = 0$ and the canonical autonomous endpoint incidence has $g = 2\hbar\nu$. Since $e = 4\mathcal{A}_{\text{ex}}^{(2)}$, either one-sided branch reduces to

$$\mathcal{A}_{\text{ex}}^{(2)} \geq \frac{\hbar\nu}{2} \frac{\text{Tr } F_N}{N}, \quad (107)$$

which is exactly the one-sided WP18 coefficient. The clean bilateral case similarly gives the $\hbar\nu/4$ coefficient.

6. Shared-kernel qutrit check

For the qutrit benchmark defined later in the supplement,

$$J_B^+ = J_B^- = \frac{5}{4}, \quad J_+ = \frac{7}{4}, \quad J_- = 3, \quad (108)$$

with

$$g_+ = g_- = \frac{13}{4}\hbar\nu, \quad 4\mathcal{A}_{\text{ex}}^{(2)} = \frac{247}{16}\hbar\nu. \quad (109)$$

In units $\hbar\nu = 1$, Eq. (101) gives

$$\Psi_{5/4}(247/16; 13/4, 13/4) = 12. \quad (110)$$

Thus the canonical autonomous reduction introduces no additional resource-layer looseness in this benchmark.

IV. MULTI-GAP SHARED-HESSIAN ACTION

A. Multi-gap shared-Hessian spectral-action sum

Let $\rho(\mathbf{x}, \mathbf{y})$ be one finite-dimensional C^2 physical family with baseline ρ_0 , support projector P , and kernel projector Q . For each positive gap ν_k , let the complex tangent be A_k and assume

$$PA_kP = 0, \quad QA_kQ = 0. \quad (111)$$

The real coordinates (x_k, y_k) are the fixed cosine/sine quadratures of A_k ; the Hessian trace below uses the corresponding direct-sum Euclidean parameter metric. No invariance under arbitrary anisotropic reparameterizations is claimed.

Define

$$X_k = QA_kP, \quad Y_k = QA_k^\dagger P, \quad (112)$$

so $A_k = X_k + Y_k^\dagger$. The support-to-kernel cosine and sine blocks are

$$K_{k,c} = \frac{X_k + Y_k}{2}, \quad K_{k,s} = \frac{X_k - Y_k}{2i}. \quad (113)$$

Second-order positivity gives

$$Q\partial_{x_k}^2\rho(0)Q \succeq 2K_{k,c}\rho_0^+K_{k,c}^\dagger, \quad (114)$$

and

$$Q\partial_{y_k}^2\rho(0)Q \succeq 2K_{k,s}\rho_0^+K_{k,s}^\dagger. \quad (115)$$

Adding cancels the mixed X_k/Y_k terms:

$$C_{\Delta,k} := Q(\partial_{x_k}^2 + \partial_{y_k}^2)\rho(0)Q \quad (116)$$

$$\succeq X_k\rho_0^+X_k^\dagger + Y_k\rho_0^+Y_k^\dagger \quad (117)$$

$$=: Z_{k,+} + Z_{k,-}. \quad (118)$$

Summing the operator inequalities before scalar pricing yields

$$\boxed{C_\Sigma := Q \sum_k (\partial_{x_k}^2 + \partial_{y_k}^2)\rho(0)Q \succeq \sum_k (Z_{k,+} + Z_{k,-}).} \quad (119)$$

Let $G \succeq 0$ be one positive kernel cost and define

$$\mathcal{A}_{G,\Sigma}^{(2)} = \frac{1}{4} \text{Tr}(GC_\Sigma). \quad (120)$$

For every nonzero orientation define $R_{k,\pm} = \text{supp } Z_{k,\pm}$ and

$$g_{k,\pm} = \lambda_{\min}(R_{k,\pm}GR_{k,\pm}|_{R_{k,\pm}}). \quad (121)$$

Equation (119) implies

$$4\mathcal{A}_{G,\Sigma}^{(2)} \geq \sum_k [g_{k,+}J_{k,+} + g_{k,-}J_{k,-}], \quad (122)$$

with absent orientations omitted and $J_{k,\pm} = \text{Tr } Z_{k,\pm}$.

For a fixed arbitrary collective POVM on N independently encoded copies, modewise score Minkowski gives

$$\sqrt{\frac{\text{Tr } F_{N,k}}{N}} \leq \sqrt{J_{k,+}} + \sqrt{J_{k,-}}. \quad (123)$$

Define the effective mode price by

$$\gamma_k = \begin{cases} (g_{k,+}^{-1} + g_{k,-}^{-1})^{-1}, & J_{k,+}, J_{k,-} > 0 \text{ and } g_{k,+}, g_{k,-} > 0, \\ g_{k,+}, & J_{k,+} > 0, J_{k,-} = 0, g_{k,+} > 0, \\ g_{k,-}, & J_{k,-} > 0, J_{k,+} = 0, g_{k,-} > 0, \\ 0, & \text{some nonzero orientation has zero restricted cost.} \end{cases} \quad (124)$$

A mode with $A_k = 0$ may simply be omitted. For positive bilateral costs, weighted Cauchy–Schwarz gives

$$\gamma_k(\sqrt{J_{k,+}} + \sqrt{J_{k,-}})^2 \leq g_{k,+}J_{k,+} + g_{k,-}J_{k,-}. \quad (125)$$

For one-sided modes the same statement is immediate, while $\gamma_k = 0$ is deliberately vacuous for a cost-null information-bearing direction. Hence

$$\boxed{\sum_k \gamma_k \frac{\text{Tr } F_{N,k}}{N} \leq 4\mathcal{A}_{G,\Sigma}^{(2)}} \quad (126)$$

If a nonzero orientation has zero restricted cost, no positive scalar information price for that mode can follow from the chosen G alone; one must change G or retain the operator-valued curvature.

1. Clean autonomous frequency weighting

Suppose each mode is an exact clock–signal exchange at gap ν_k and the two endpoint ranges are mode separated. Choose G to assign combined clock-plus-signal incidence price

$$2\hbar\nu_k \quad (127)$$

to either orientation of mode k . Then $g_{k,+} = g_{k,-} = 2\hbar\nu_k$ for bilateral modes and

$$\gamma_k = \hbar\nu_k. \quad (128)$$

Equation (126) becomes

$$\boxed{\mathcal{A}_{G,\Sigma}^{(2)} \geq \sum_k \frac{\hbar\nu_k}{4} \frac{\text{Tr } F_{N,k}}{N}} \quad (129)$$

2. One common sharp Fourier measurement

Take

$$|n\rangle = |m-n\rangle_C |m+n\rangle_S, \quad n = -m, \dots, m, \quad (130)$$

with baseline $|0\rangle$. For $k = 1, \dots, m$ choose

$$A_k = c_k(|k\rangle\langle 0| + |0\rangle\langle -k|), \quad \nu_k = k\omega_0. \quad (131)$$

The exact common family is

$$|\psi(\mathbf{x}, \mathbf{y})\rangle = \sqrt{1 - \frac{1}{2} \sum_{k=1}^m c_k^2 (x_k^2 + y_k^2)} |0\rangle \quad (132)$$

$$+ \sum_{k=1}^m \frac{c_k}{2} (x_k + iy_k) |-k\rangle + \sum_{k=1}^m \frac{c_k}{2} (x_k - iy_k) |k\rangle. \quad (133)$$

It lies entirely in one total-energy eigenspace and is physical on

$$\frac{1}{2} \sum_k c_k^2 (x_k^2 + y_k^2) < 1. \quad (134)$$

Its kernel Hessian is

$$C_\Sigma = \sum_{k=1}^m c_k^2 (|-k\rangle\langle -k| + |k\rangle\langle k|). \quad (135)$$

Let $d = 2m + 1$ and measure

$$|v_j\rangle = \frac{1}{\sqrt{d}} \sum_{n=-m}^m e^{in\phi_j} |n\rangle, \quad \phi_j = \frac{2\pi j}{d}. \quad (136)$$

At the baseline $p_j = 1/d$ and

$$z_{j,k} = \langle v_j | A_k | v_j \rangle = \frac{2c_k}{d} e^{-ik\phi_j}. \quad (137)$$

Discrete harmonic orthogonality gives

$$F_{x_k x_l} = 2c_k^2 \delta_{kl}, \quad (138)$$

$$F_{y_k y_l} = 2c_k^2 \delta_{kl}, \quad (139)$$

$$F_{x_k y_l} = 0. \quad (140)$$

Thus $\text{Tr } F_{1,k} = 4c_k^2$ for every mode simultaneously. With

$$G = 2\hbar\omega_0 \sum_{n \neq 0} |n| |n\rangle\langle n|, \quad (141)$$

$$\mathcal{A}_{G,\Sigma}^{(2)} = \sum_k \hbar\nu_k c_k^2 = \sum_k \frac{\hbar\nu_k}{4} \text{Tr } F_{1,k}. \quad (142)$$

Hence one common measurement simultaneously saturates every mode and the complete weighted sum.

For overlapping ranges, Eq. (126) remains valid for any supplied $G \succeq 0$, but a unique frequency-diagonal interpretation is not claimed.

V. EXACT COMMON-RECORD ACCESSIBILITY BENCHMARK

A. Shared-kernel qutrit benchmark

Use a three-level fixed-total-energy shell with ordered basis $\{|0\rangle, |1\rangle, |2\rangle\}$ and define

$$|q\rangle = \frac{1}{2}|0\rangle + \sqrt{\frac{5}{8}}|1\rangle + \frac{1}{2\sqrt{2}}|2\rangle. \quad (143)$$

Let

$$Q = |q\rangle\langle q|, \quad P = \mathbb{I} - Q, \quad \rho_0 = \frac{P}{2}, \quad (144)$$

and choose the exact positive-gap tangent

$$A = |1\rangle\langle 0| - \sqrt{2}|2\rangle\langle 1|. \quad (145)$$

An orthonormal basis of P is

$$|e_1\rangle = \frac{1}{\sqrt{3}}|0\rangle - \sqrt{\frac{2}{3}}|2\rangle, \quad (146)$$

$$|e_2\rangle = -\frac{\sqrt{15}}{6}|0\rangle + \frac{\sqrt{6}}{4}|1\rangle - \frac{\sqrt{30}}{12}|2\rangle. \quad (147)$$

In the ordered basis $\{|e_1\rangle, |e_2\rangle, |q\rangle\}$,

$$\rho_0 = \text{diag}(1/2, 1/2, 0) \quad (148)$$

and

$$A = \begin{pmatrix} 0 & \sqrt{2}/2 & \sqrt{30}/6 \\ \sqrt{2}/4 & 0 & \sqrt{6}/3 \\ \sqrt{30}/12 & -\sqrt{6}/3 & 0 \end{pmatrix}. \quad (149)$$

For the support/kernel decomposition $B = PAP$, $K_+ = QAP$, $K_- = QA^\dagger P$, the exact weighted norms are

$$J_B^+ = J_B^- = \frac{5}{4}, \quad J_+ = \frac{7}{4}, \quad J_- = 3. \quad (150)$$

The minimal shared kernel curvature is

$$C_\Delta = \frac{19}{4}Q. \quad (151)$$

For the autonomous embedding, the canonical endpoint-incidence operator has physical-basis representation

$$\frac{G_{\text{ex}}}{\hbar\nu} = \text{diag}(2, 4, 2). \quad (152)$$

Hence both synthesized rank-one ranges have restricted price

$$g_+ = g_- = \frac{13}{4}\hbar\nu, \quad (153)$$

and the minimal-curvature action obeys

$$4\mathcal{A}_{\text{ex}}^{(2)} = \frac{247}{16}\hbar\nu. \quad (154)$$

With $a_+ = a_- = 5/4$, the mixed resource envelope returns exactly

$$\Psi_{5/4}(247/16; 13/4, 13/4) = 12 \quad (155)$$

in units $\hbar\nu = 1$.

The two-quadrature SLD-QFI matrix for the same baseline tangent is

$$F_Q = \text{diag}\left(\frac{39}{8}, \frac{47}{8}\right), \quad \text{Tr } F_Q = \frac{43}{4}. \quad (156)$$

The following subsection proves that the exact one-copy common-record classical Fisher supremum is only $55/8$. Thus this single finite-dimensional model separates physical resource, SLD tangent geometry, and accessible common-record information.

B. Exact common-record Fisher supremum for the shared-kernel qutrit

Use the fixed-total-excitation qutrit benchmark defined in the main supplement. Choose an orthonormal basis $\{|e_1\rangle, |e_2\rangle\}$ for the support of ρ_0 such that, in the ordered basis $\{|e_1\rangle, |e_2\rangle, |q\rangle\}$,

$$\rho_0 = \text{diag}(1/2, 1/2, 0) \quad (157)$$

and

$$A = \begin{pmatrix} 0 & \sqrt{2}/2 & \sqrt{30}/6 \\ \sqrt{2}/4 & 0 & \sqrt{6}/3 \\ \sqrt{30}/12 & -\sqrt{6}/3 & 0 \end{pmatrix}. \quad (158)$$

For a POVM effect M_y , let

$$p_y = \text{Tr}(\rho_0 M_y), \quad z_y = \text{Tr}(A M_y). \quad (159)$$

The two-quadrature classical Fisher trace is

$$\text{Tr } F_1 = \sum_{y: p_y > 0} \frac{|z_y|^2}{p_y}. \quad (160)$$

a. Rank-one refinement. If $M = \sum_j M_j$ is decomposed into positive effects and $p_j = \text{Tr}(\rho_0 M_j)$, $z_j = \text{Tr}(A M_j)$, then quadratic-over-linear convexity gives

$$\frac{|\sum_j z_j|^2}{\sum_j p_j} \leq \sum_j \frac{|z_j|^2}{p_j}. \quad (161)$$

Therefore refining effects into rank-one components cannot decrease the Fisher trace. It is sufficient to establish a pointwise bound for $M = w|\phi\rangle\langle\phi|$.

b. Dual quadratic witness. Define

$$Y = \begin{pmatrix} 9/16 & 0 & 0 \\ 0 & 9/16 & 3\sqrt{15}/8 \\ 0 & 3\sqrt{15}/8 & 23/4 \end{pmatrix}. \quad (162)$$

We claim

$$|\langle\phi|A|\phi\rangle|^2 \leq \langle\phi|\rho_0|\phi\rangle\langle\phi|Y|\phi\rangle \quad (163)$$

for every vector $|\phi\rangle$.

For $\lambda > 0$ and phase θ , define

$$\mathcal{M}(\lambda, \theta) = \lambda\rho_0 + \lambda^{-1}Y - (e^{i\theta}A + e^{-i\theta}A^\dagger). \quad (164)$$

Set $x = \lambda^2$ and $t = \cos^2\theta$. Direct symbolic reduction gives the first leading principal minor

$$\frac{8x+9}{16\lambda} > 0, \quad (165)$$

the second leading principal minor

$$\frac{t(8x-9)^2 + (1-t)(64x^2 + 112x + 81)}{256x}, \quad (166)$$

and the determinant

$$\frac{t(8x-9)^2 + (1-t)(40x+81)}{128\lambda^3}. \quad (167)$$

For $x > 0$ and $0 \leq t \leq 1$, both numerators are manifestly nonnegative. They are strictly positive except at the isolated point $t = 1$, $x = 9/8$; positive semidefiniteness there follows by continuity. Hence

$$\mathcal{M}(\lambda, \theta) \succeq 0 \quad \text{for all } \lambda > 0, \theta \in \mathbb{R}. \quad (168)$$

For an arbitrary $|\phi\rangle$, choose θ to align the phase of $\langle\phi|A|\phi\rangle$. Then

$$2|\langle\phi|A|\phi\rangle| \leq \lambda\langle\phi|\rho_0|\phi\rangle + \lambda^{-1}\langle\phi|Y|\phi\rangle. \quad (169)$$

Minimizing over λ proves Eq. (163).

For a rank-one POVM, Eq. (163) gives

$$\text{Tr } F_1 \leq \sum_y w_y \langle \phi_y | Y | \phi_y \rangle = \text{Tr } Y = \frac{55}{8}. \quad (170)$$

Rank-one refinement extends the upper bound to every POVM.

c. Sharp limiting projective sequence. Write Eq. (158) in block form

$$A = \begin{pmatrix} B & b \\ a^\dagger & 0 \end{pmatrix}, \quad (171)$$

where

$$B = \begin{pmatrix} 0 & \sqrt{2}/2 \\ \sqrt{2}/4 & 0 \end{pmatrix}, \quad a = \begin{pmatrix} \frac{\sqrt{30}}{12}, -\frac{\sqrt{6}}{3} \end{pmatrix}, \quad b = \begin{pmatrix} \frac{\sqrt{30}}{6} \\ \frac{\sqrt{6}}{3} \end{pmatrix}. \quad (172)$$

The numerical radius of B is

$$w(B) = \frac{3\sqrt{2}}{8}. \quad (173)$$

The two orthogonal support vectors

$$|s_\pm\rangle = \frac{|e_1\rangle \pm |e_2\rangle}{\sqrt{2}} \quad (174)$$

therefore contribute a total

$$\frac{9}{8} \quad (175)$$

to the Fisher trace.

For a vector approaching the kernel,

$$|\phi_\epsilon\rangle = \cos \epsilon |q\rangle + \sin \epsilon |t\rangle, \quad (176)$$

the baseline probability is $\sin^2 \epsilon/2$, while its Fisher contribution tends to

$$2|\langle q|A|t\rangle + \langle t|A|q\rangle|^2. \quad (177)$$

Choosing

$$|t_*\rangle = i \frac{(a-b)^T}{\|a-b\|} \quad (178)$$

gives

$$\|a-b\|^2 = \frac{23}{8}, \quad (179)$$

so the near-kernel contribution tends to

$$\frac{23}{4}. \quad (180)$$

Let U_ϵ be a unitary rotation by angle ϵ in $\text{span}\{|q\rangle, |t_*\rangle\}$ and the identity on its orthogonal complement. The projective measurement

$$\{U_\epsilon|s_+\rangle, U_\epsilon|s_-\rangle, U_\epsilon|q\rangle\} \quad (181)$$

has ordinary positive baseline probabilities for every sufficiently small $\epsilon > 0$, and

$$\lim_{\epsilon \rightarrow 0^+} \text{Tr } F_1[U_\epsilon] = \frac{9}{8} + \frac{23}{4} = \frac{55}{8}. \quad (182)$$

Combining Eqs. (170) and (182),

$$\boxed{\sup_{\text{POVM}} \text{Tr } F_1 = \frac{55}{8}.} \quad (183)$$

The result is a supremum; exact attainment by a regular POVM is not required.

For this benchmark the three distinct layers are therefore

$$12 > \frac{43}{4} > \frac{55}{8}, \quad (184)$$

corresponding respectively to the physical shared-curvature resource ceiling, the SLD-QFI trace, and the exact accessible one-copy common-record Fisher supremum.

VI. INTERPRETIVE SEPARATION OF RESOURCE, QFI, AND ACCESSIBILITY

The inequalities proved above are physical necessary-resource bounds on an encoded state family. They are not SLD-attainability statements. Conversely, a QFI matrix is a quantum-statistical tangent bound and need not equal the physical resource ceiling. The shared-kernel qutrit makes this separation explicit:

$$12 > \frac{43}{4} > \frac{55}{8}, \quad (185)$$

where the three quantities are respectively the physical resource ceiling, the SLD-QFI trace, and the exact one-copy common-record Fisher supremum. This distinction is important

at rank-changing boundaries, where zero-probability directions and incompatible score quadratures can separate the three layers substantially [7, 8].

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